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PHYSICS

1) An astronomical telescope and galilean telescope use identical objective lenses. They have the same magnification when both are in normal adjustment. The eyepiece of the astronomical telescope has a focal length f, then :-

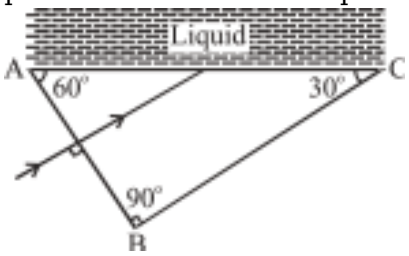
(1) Tube lengths of two telescopes differ by f

(2) Tube lengths of two telescopes differ by 2f

(3) f_e for galilean telescope is twice that of astronomical telescope

(4) Galilean telescope has longer tubelength

2) Light is incident normally on face AB of a prism as shown in figure. A liquid of refractive index μ is placed on face AC of the prism. The prism is made of glass of refractive index 3/2 :-



The limits of μ for which total internal reflection takes place on face AC is :

(1) $\mu > \frac{\sqrt{3}}{2}$

(2) $\mu < \frac{3\sqrt{3}}{4}$

(3) $\mu > \sqrt{3}$

(4) $\mu < \sqrt{3}$

3) A small piece of wire, bent into L shape with upright and horizontal portions of equal lengths, is placed with the horizontal portion along the axis of the concave mirror whose radius of curvature is 10 cm. If the bend is 20 cm from the pole of the mirror, then the ratio of the lengths of the images of the upright and horizontal portions of the wire is

(1) 1 : 2

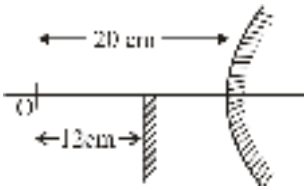
(2) 3 : 1

(3) 1 : 3

(4) 2 : 1

4) A luminous object is placed at a distance of 20 cm from convex mirror. A plane mirror is set so that virtual images formed by both mirror will coincide. If plane mirror is at distance of 12 cm from

object, find out the focal length of convex mirror :-



(1) + 5 cm

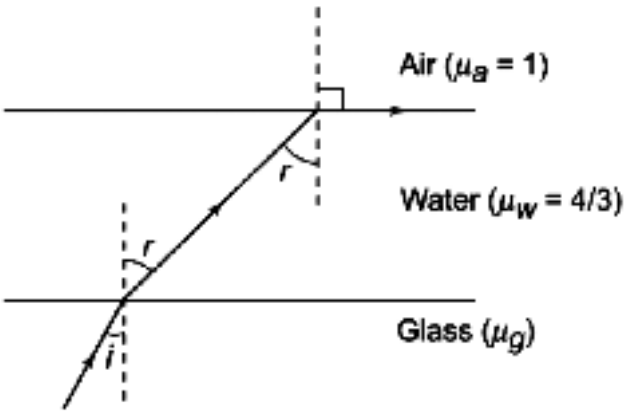
(2) – 5 cm

(3) 15 cm

(4) 25 cm

5) **Statement-1 :** A ray of light is incident at the galss-water interface at an angle i. It emerges finally parallel to the surface of water as shown in figure. Then $\mu_g = \left(\frac{4}{3}\right) \sin i$

Statement-2 : The refracted ray follows a path in accordance with Snell's law of refraction.



(1) Statement-1 is True, Staternent-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(3) Statement-1 is True, Staternent-2 is False.

(4) Statement-1 is False, Statement-2 is True.

6) If the focal length of objective and eye lens are 1.2 cm and 3 cm respectively and the object is put 1.25 cm away from the objective lens and the final image is formed at infinity. The magnifying power of the microscope is :-

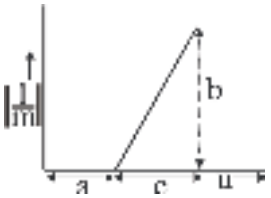
(1) 150

(2) 200

(3) 250

(4) 400

7) The graph shows how the reciprocal of magnification produced by a converging thin lens varied with object distance u in displacement method. What was the focal length of the lens used ?



(1) a/b

(2) bc/a

(3) c/b

(4) ab/c

8) A hollow concave lens is made of very thin transparent material. It can be filled with air or either of two liquids L_1 or L_2 having refractive indices n_1 and n_2 respectively ($n_2 > n_1 > 1$). The lens will diverge a parallel beam of light if it is filled with

(1) air and placed in air

(2) air and immersed in L_1

(3) L_1 and immersed in L_2

(4) L_2 and immersed in L_1

9) Two convex lenses of focal lengths f_1 and f_2 are mounted coaxially separated by certain distance. If the power of the combination is zero, the distance between the lenses is

(1) $|f_1 - f_2|$

(2) $f_1 + f_2$

(3) $\frac{f_1 f_2}{|f_1 - f_2|}$

(4) $\frac{f_1 f_2}{f_1 - f_2}$

10) When a lens of power P (in air) made of material of refractive index μ is immersed in liquid of refractive index μ_0. Then the power of lens is:

(1) $\frac{\mu - 1}{\mu - \mu_0} P$

(2) $\frac{\mu - \mu_0}{\mu - 1} P$

(3) $\frac{\mu - \mu_0}{\mu - 1} \cdot \frac{P}{\mu_0}$

(4) none of these

11) **Assertion (A) :-** A lens of positive power will convert to a lens of negative power when placed in a medium of refractive index higher than refractive index of material of lens. **Reason (R) :-** Power of lens depends on difference of refractive index of lens and medium.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is incorrect but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

12) A thin lens has focal length f and its small aperture has diameter d. It forms an image of intensity I. Now, the central part of the aperture up to diameter (d/2) is blocked by an opaque paper. Now, the focal length and image intensity will :-

(1) (f/2) and (I/2)

(2) f and (I/4)

(3) (3f/4) and (I/2)

(4) f and (3I/4)

13) The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside the lens is $\frac{2}{3}$ times the wavelength in free space. The radius of the curved surface of the lens is :-

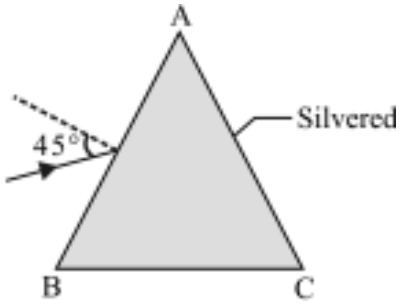
(1) 1 m

(2) 2 m

(3) 3 m

(4) 6 m

14) A prism ABC of angle 30° at A has its face AC silvered. A ray of light incident at an angle of 45° at the face AB retraces its path after refraction at face AB and reflection at face AC. The refractive index of the material of the prism is :-



(1) 1.5

(2) $\frac{3}{\sqrt{2}}$

(3) $\sqrt{2}$

(4) $\frac{4}{3}$

15) A fish rising vertically up towards the surface of water with speed 3m/s observes a bird diving vertically down towards it with speed 9 m/s. The actual velocity of bird is:

(1) 4.5 m/sec

(2) 5.4 m/sec

(3) 3.0 m/sec

(4) 3.4 m/sec

16) An air bubble is situated in glass slab of length 25 cm. When it is seen from one side, it appears at 6 cm and when seen from other side, it appears at 4 cm. Find refractive index of slab :-

(1) 1.5

(2) 2.5

(3) 1

(4) 1.33

17) The thickness of a slab is $5\sqrt{3}$ cm and refractive index $\sqrt{3}$. If light incident on it at angle of incidence 60°, then find lateral shift between direction of incident ray and emergent ray produced by it :-

(1) $\frac{5\sqrt{3}}{2}$cm

(2) $\frac{5}{2}$cm

(3) $\frac{5}{\sqrt{3}}$cm

(4) 5 cm

18) A point object is moving on the principal axis of a concave mirror of focal length 24 cm towards the mirror. When it is at a distance of 60 cm from the mirror, its velocity is 9 cm/sec. What is the velocity of the image at that instant -

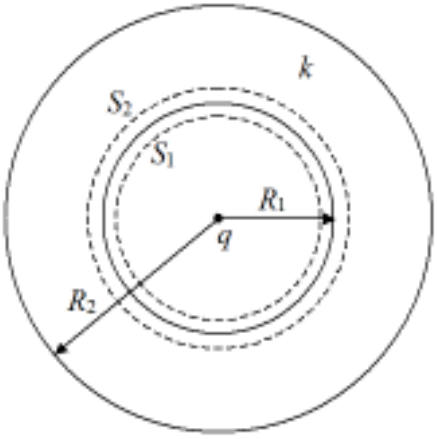
(1) 5 cm/sec

(2) 12 cm/sec

(3) 4 cm/sec

(4) 9 cm/sec

19) A positive charge q is kept at the center of a thick shell of inner radius R_1 and outer radius R_2 which is made up of conducting material. If ϕ_1 is flux through closed gaussian surface S_1 whose radius is just less than R_1 and ϕ_2 is flux through closed gaussian surface S_2 whose radius is just greater than R_1 then:-



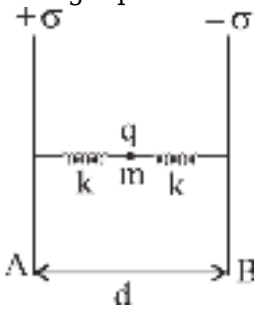
(1) $\phi_1 > \phi_2$

(2) $\phi_2 > \phi_1$

(3) $\phi_1 = \phi_2 = \frac{q}{\varepsilon_0}$

(4) $\phi_1 = \phi_2 = \frac{kq}{\varepsilon_0}$

20) Two large conducting plates having surface charge densities + σ and – σ respectively are fixed 'd' distance apart. A small test charge q of mass m is attached to two identical springs as shown in the adjacent figure. The charge q is now released from rest with springs in natural length. Then q will : [neglect gravity] –



(1) perform SHM with angular frequency $\sqrt{\frac{2k}{m}}$

(2) perform SHM with amplitude $\frac{q}{2k\varepsilon_0\sigma}$

(3) not perform SHM but will have a periodic motion

(4) remain stationary

21) The charges on two parallel copper plates separated by a small distance are +Q and –Q. A test charge q experiences a force F when placed between these plates. Now if one of the plates is removed to infinity, then the force on the test charge will become–

(1) F/2

(2) F

(3) 2F

(4) zero

22) Four charges 2C, – 3C, –4C and 5C respectively are placed at the four corners of a square. Which of the following statements is true for the point of intersection of the diagonals ?

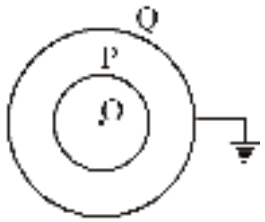
(1) E = 0, V = 0

(2) E ≠ 0, V = 0

(3) E = 0, V ≠ 0

(4) E ≠ 0, V ≠ 0

23) P and Q are two concentric metallic shells. If P is positively charged and Q is earthed. then:-



(1) charge density on Q is same as on P

(2) electric field between P & Q is uniform

(3) electric potential inside P is zero

(4) electric field outside Q is zero

24) If electric potential of an electric field is given by $V = 3x^2 - y^2 + 2z^2$ volt than find electric field at point (1,1,1) m

(1) $\sqrt{56}$ V/m

(2) $\sqrt{148}$ V/m

(3) 6 V/m

(4) $\sqrt{29}$ V/m

25) Match the column

	Column-I		**Column-II**
(a)	Electric field outside a conducting charged sphere	(e)	Constant
(b)	Electric potential outside a conducting charged sphere	(f)	directly proportional to distance from centre
(c)	Electric field inside a non-conducting charged sphere	(g)	inversely proportional to distance from center
(d)	Electric potential inside a charged conducting sphere	(h)	inversely proportional to the square of distance from center

(1) a - h, b - g, c - e, d - f

(2) a - e, b - f, c - h, d - g

(3) a - h, b - g, c - f, d - e

(4) a - g, b - h, c - f, d – e

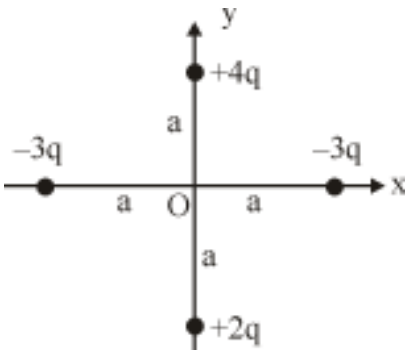
26) **Assertion :** If a proton and an electron are placed in the same uniform electric field they experience different acceleration
Reason : Electric force on a test charge is independent of its mass.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.



27) Dipole moment of the system of charges is

(1) $qa(+\hat{i})$

(2) $2qa(-\hat{i})$

(3) $2qa(+\hat{j})$

(4) $-2qa(+\hat{j})$

28) Four charges equal to –Q are placed at the four corners of a square and a charge q at a centre. If the system is in equilibrium, then the value of q is

(1) $-\frac{Q}{4}(1+2\sqrt{2})$

(2) $\frac{Q}{4}(1+2\sqrt{2})$

(3) $-\frac{Q}{2}(1+2\sqrt{2})$

(4) $\frac{Q}{2}(1+2\sqrt{2})$

29) A particle of mass 2kg and charge 1mC is projected vertically with a velocity 10 m/s. There is a uniform horizontal electric field of 10^4 N/C. ($g = 10$ m/s^2)
(a) The horizontal range of particle is 10m.
(b) The time of flight of particle is 2 seconds.
(c) The maximum height reached is 5m.
(d) The horizontal range of particle is 5m.

(1) a,b are correct

(2) b,c are correct

(3) b,c,d are correct

(4) a,b,c are correct

30) 1000 drops of same size are charged to a potential of 1V each. If they coalesce to form a single drop, its potential would be.

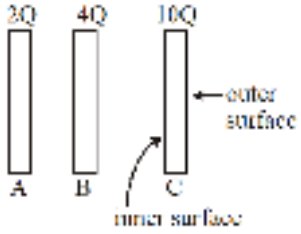
(1) V

(2) 10V

(3) 100V

(4) 1000V

31) Three very large metal plates are given charges as shown in the figure. If cross sectional area of



each plate is same then final charge distribution on plate C is :

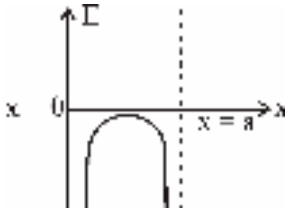
(1) +5Q on inner surface and +5Q on outer surface

(2) +6Q on inner surface and +4Q on outer surface

(3) +7Q on inner surface and +9Q on outer surface

(4) +8Q on inner surface and +2Q on outer surface

32) Two point charges Q_1, Q_2 are fixed at x = 0 and x = a. Assuming that field strength is positive in



the direction coinciding with the positive direction of x. then, which following option will be correct ?

(1) Both Q_1 and Q_2 are negative with $|Q_1| > |Q_2|$

(2) Q_1 is positive and Q_2 is negative with $|Q_1| > |Q_2|$

(3) Q_1 is negative and Q_2 is positive with $|Q_1| = |Q_2|$

(4) Both are positive $|Q_1| > |Q_2|$

33) **Statement-A :** Gauss law of electric flux is valid because coulomb's law of force is inverse square law.

Statement-B : Electric field in the vicinity of charge conductor of irregular shape is always $\frac{\sigma}{\in_0}\hat{n}$, where σ is local surface charge density and $\hat{n}$ is unit vector normal to conductor at that point.

Statement-C : In gauss law of flux over closed surface $\oint \vec{E}.\vec{ds} = \frac{\Sigma q}{\in_0}$, here electric field E and Σq is due to same charge.

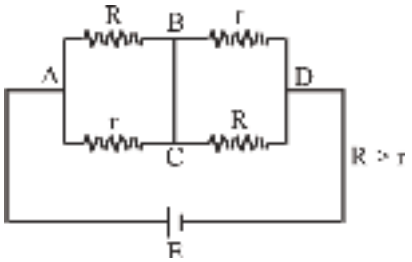
Which of the following statement is correct is.

(1) c is correct and a and b are incorrect

(2) a and c are correct and b is incorrect

(3) a, b and c all are correct

(4) a and b are correct and c is incorrect.



34) For given circuit, select correct statement

Equivalent resistance between A and D is

(1) $\frac{2Rr}{R+r}$

(2) Current through BC is $\frac{E(R-r)}{2rR}$

(3) Current through BC is from C to B

(4) All of these

35) A galvanometer has a coil of resistance 100Ω and shows a full scale deflection, when 100μA current is passed through it. Resistance required to use it as a voltmeter of range 5V, will be:

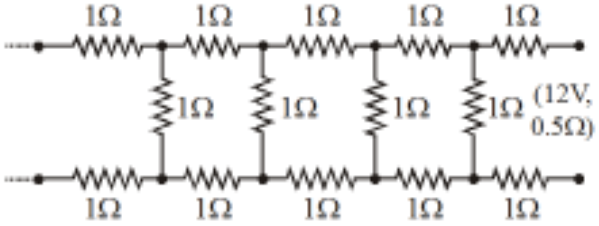
(1) 95 kΩ

(2) 9.5 kΩ

(3) 49.9 kΩ

(4) 4.99 kΩ.

36) Determine the current drawn from a 12V supply with internal resistance 0.5Ω by the infinite network shown in given figure. Each resistor has 1Ω resistance :



(1) 3 A

(2) 2.7 A

(3) 3.7 A

(4) 4.2 A

37) A voltmeter of resistance 1000Ω gives full scale deflection when a current of 100 mA flow through it. The shunt resistance required across it to enable it to be used as an ammeter reading 1 A at full scale deflection is

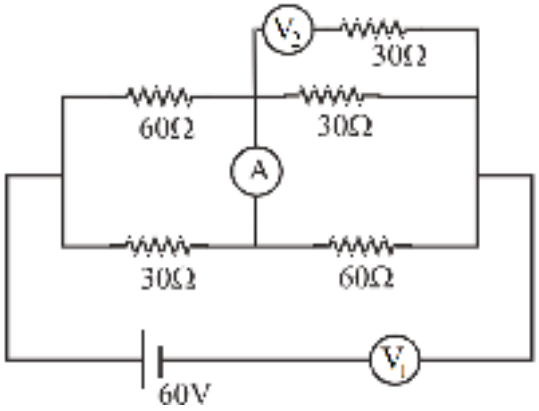
(1) 10000 Ω

(2) 9000 Ω

(3) 222 Ω

(4) 111Ω

38) Find readings of voltmeters V_1 and V_2 and ammeter in given circuit –



(resistance of V_1 is 80Ω and V_2 and A are ideal):-

(1) $V_1 = 42.8$ V, $V_2 = 4.2$ V, $i = \frac{11}{60}A$

(2) $V_1 = 30$ V, $V_2 = 15$ V, $i = \frac{11}{60}A$

(3) $V_1 = 40$ V, $V_2 = 10$ V, $i = \frac{1}{6}A$

(4) $V_1 = 40$ V, $V_2 = 15$ V, $i = \frac{11}{60}A$

39) Two cities are 150 km apart. Electric power is sent from one city to another city through copper wires. The fall of potential per km is 8V and the average resistance per km is 0.5 Ω. The power loss in the wire is :-

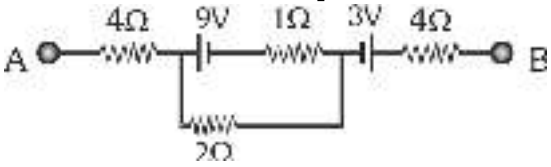
(1) 19.2 W

(2) 19.2 kW

(3) 19.2 MW

(4) 12.2 kW

40) In the circuit shown in figure potential difference between point A and B is 16V. Find the current passing through 2Ω resistance.



(1) 3.5 A

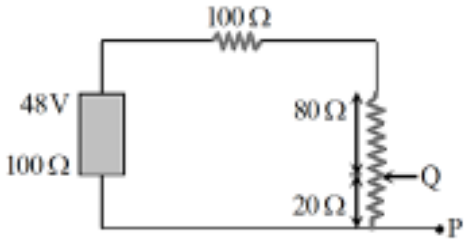
(2) 0 A

(3) 2.5 A

(4) 4 A

41)

In the circuit, the potential difference across PQ will be nearest to



(1) 9.6 V

(2) 6.6 V

(3) 4.8 V

(4) 3.2 V

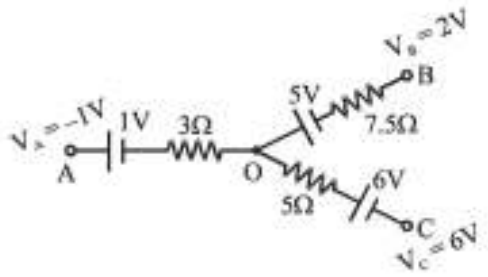
42) A non-conducting ring of radius R has two positive point charges lying diametrically opposite to each other, each of magnitude Q. The ring rotates with an angular velocity ω about it's axis. If I is the equivalent current then,

(1) $I = Q\omega$

(2) $I = \frac{2Q\omega}{\pi}$

(3) $I = \frac{Q\omega}{\pi}$

(4) I = Zero

43) 

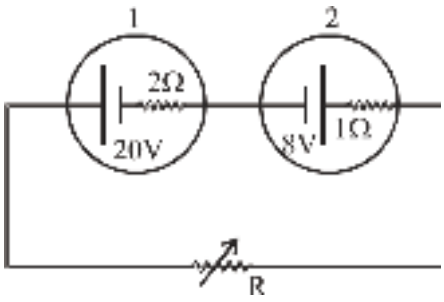
In the above diagram :-

(1) Potential at junction 'O' is zero

(2) Current in 3Ω is 2A

(3) Current in 5 Ω is 2.3 A

(4) All the above are correct

44) 

When power in variable resistance is maximum, then find value of R :-

(1) 2 Ω

(2) 1 Ω

(3) 4 Ω

(4) 3 Ω

45) An unknown resistance R_1 is connected in series with a resistance of 10 Ω. This combination is connected to one gap of a metre bridge, while a resistance R_2 is connected in the other gap. The balance point is at 50 cm. Now when the 10 Ω resistance is removed, the balance point shifts to 40 cm. The value of R_1 is :-

(1) 10 Ω

(2) 20 Ω

(3) 40 Ω

(4) 60 Ω

CHEMISTRY

1) For the gas phase reaction :
$2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$, $\Delta H = -43.5$ kCal mol^{-1},
which one of the statement below is true for $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$:-

(1) K is independent of temperature

(2) K increases as temperature decreases

(3) K decreases as temperature decreases

(4) K varies with the addition of NO

2) Match Column–I (hypothetical reactions) with Column–II (ratio $\frac{K_P}{K_C}$ for the given reactions) and select the correrct answer using the code given below the lists :-

	Column–I		**Column–II**
(A)	$A_2(g) + 3B_2(g) \rightleftharpoons 2AB_3(g)$	(P)	$(RT)^{-2}$
(B)	$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$	(Q)	$(RT)^{0}$
(C)	$A(s) + 1.5\ B_2(g) \rightleftharpoons AB_3(g)$	(R)	$(RT)^{\frac{1}{2}}$
(D)	$AB_2(g) \rightleftharpoons AB(g) + 0.5\ B_2(g)$	(S)	$(RT)^{-\frac{1}{2}}$

(1) (A)–(S), (B)–(R), (C)–(Q), (D)–(P)

(2) (A)–(P), (B)–(S), (C)–(R), (D)–(Q)

(3) (A)–(P), (B)–(Q), (C)–(S), (D)–(R)

(4) (A)–(R), (B)–(P), (C)–(Q), (D)–(S)

3) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** :
Assertion (A) :-
$Fe^{+3}(aq.) + SCN^{-}(aq.) \rightleftharpoons [Fe(SCN)]^{2+}(aq.)$
yellow colourless deep red
On addition of oxalic acid the intensity of red colour decreases.
Reason (R) :- Oxalic acid ($H_2C_2O_4$) reacts with Fe^{+3} ions to form stable complex $[Fe(C_2O_4)_3]^{3-}$

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

4) In the reaction $C_{(s)} + CO_{2(g)} \rightleftharpoons 2CO_{(g)}$ the equilibrium pressure is 12 atm. If 50% of CO_2 reacts then K_p will be :-

(1) 12 atm

(2) 16 atm

(3) 20 atm

(4) 24 atm

5) For gaseous homogeneous equilibrium
$A(g) + 2B(g) \rightleftharpoons 3C(g) + D(g)$

When 4 moles of A and 7 moles of B was taken in a container of volume 2L. At equilibrium $\frac{[C]}{[B]} = 2$, then concentration of C at equilibrium will be:-

(1) 2 M

(2) 3 M

(3) 6 M

(4) 14 M

6) Densities of diamond and graphite are 3.5 and 2.3 g/mL. C (diamond) $\rightleftharpoons$ C (graphite) $\Delta_r H = -1.9$ kJ/mole
favourable conditions for formation of diamond are

(1) high pressure and low temperature

(2) low pressure and high temperature

(3) high pressure and high temperature

(4) low pressure and low temperature

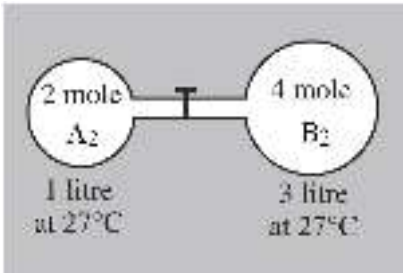
7) PQ_2 dissociates as -
$PQ_2(g) \rightleftharpoons PQ(g) + Q(g)$
Initial pressure of PQ_2 is 600 mm Hg. At equilibrium the total pressure is 800 mm Hg. Calculate the value of K_P :-

(1) 100 atm^2

(2) 400 atm

(3) 200 atm

(4) 100 atm

8) 

The gas A_2 in the left flask allowed to react with gas B_2 present in right flask as $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$; Kc = 4 at 27°C. What is the concentration of AB when equilibrium is established?

(1) 1.33 M

(2) 2.66 M

(3) 0.66 M

(4) 0.33 M

9) For $I_2(g) \rightleftharpoons 2I(g)$, K_C at 1000 K is 10^{-6}. One mole I_2 is added in one litre container. Which of the following expression is correct for the above system at equilibrium :-

(1) $[I_2] + [I] = 1 - x$

(2) $[I_2(g)] >>> [I(g)]$

(3) Both (1) & (2)

(4) $[I_2] = \frac{1}{2}[I]$

10) $\log\frac{K_P}{K_C} + \log RT = 0$ is a true relationship for the gaseous reaction :

(1) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

(2) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

(3) $H_2 + I_2 \rightleftharpoons 2HI$

(4) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

11)

Match the following for reaction

$CH_3COOH + NH_3 \rightleftharpoons CH_3COO^{\ominus} + \overset{\oplus}{N}H_4$

	Column-I		**Column-II**
(A)	K_a of CH_3COOH × K_b for $CH_3COO^{\ominus}$ at 25°C	(P)	10^{-14}
(B)	Conjugate acid of NH_3	(Q)	$> 10^{-14}$
(C)	Ionic product of water at 90ºC	(R)	$CH_3COO^{\ominus}$
(D)	Conjugate base of CH_3COOH	(S)	$NH_2^{\ominus}$

		(T)	$NH_4^{\oplus}$

(1) A→P, B→S, C→Q, D→R

(2) A→Q, B→S, C→P, D→R

(3) A→U, B→S, C→P, D→R

(4) A→P, B→T, C→Q, D→R

12) **Assertion :** Addition of HCl(aq) to HCOOH(aq), decrease the ionization of HCOOH(aq)
Reason : Due to common ion effect of H^+, ionization of HCOOH decreased.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false

13) Which of the following statement(s) is/are correct.
(A) The pH of 1×10^{-8} M HCl is 8
(B) The conjugate base of $H_2PO_4^-$ is HPO_4^{2-}
(C) k_w increases with increase in temperature
(D) When a solution of weak monoprotic acid is titrated against a strong base at half neutralisation point
 pH = $\frac{1}{2}$ pKa.

(1) A, D

(2) B, C

(3) A, B, C

(4) B, C, D

14) Calculate the minimum concentration of sulphate ion required to precipitate $BaSO_4$ in solution containing 10^{-4} M of Ba^{+2}. (K_{SP} $BaSO_4 = 4 \times 10^{-10}$)

(1) 4×10^{-6}

(2) 4×10^{-10}

(3) 2×10^{-6}

(4) 2×10^{-10}

15) A saturated solution of $CaCl_2$ has concentration of Cl^- ion 2×10^{-6} M. Find solubility of $CaCl_2$ in presence of 10^{-2} M $BaCl_2$ solution :-

(1) 10^{-14}

(2) 2×10^{-6}

(3) 4×10^{-14}

(4) None of the above

16)

Following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations :

a. $60mL\frac{M}{10}HCl + 40mL\frac{M}{10}NaOH$

b. $55mL\frac{M}{10}HCl + 45mL\frac{M}{10}NaOH$

c. $75mL\frac{M}{5}HCl + 25mL\frac{M}{5}NaOH$

d. $100mL\frac{M}{10}HCl + 100mL\frac{M}{10}NaOH$

pH of which one of them will be equal to 1 ?

(1) b

(2) a

(3) d

(4) c

17) The solubility of salt in saturated aqueous solution of NH_4KHPO_3 is:-

(1) $(K_{sp})^{1/3}$

(2) $\left(\frac{K_{sp}}{4}\right)^{1/3}$

(3) $(K_{sp})^{1/4}$

(4) $\left(\frac{K_{sp}}{4}\right)^{1/4}$

18) An acid solution with pH = 6 at 25°C is diluted by 10^2 times. The pH of solutions will :-

(1) Decrease by 2

(2) Increase by 2

(3) Decrease by 0.95 approximately

(4) Increase by 0.95 approximately

19) 200 ml of an aqueous solution of HCl (pH=2) is mixed with 300 ml of an aqueous solution of NaOH (pH = 12). Thus, pH of the resultant mixture is :-

(1) 14.0

(2) 10.2

(3) 7.9

(4) 11.3

20) K_a for CH_3COOH is 1.8×10^{-5}. Find out the percentage dissociation of 0.2M CH_3COOH in 0.1M HCl solution

(1) 0.018 %

(2) 0.36 %

(3) 18 %

(4) 36 %

21) $A^{-n_2} \rightarrow A^{-n_1} + xe^-$; then value of x is :

(1) $(n_1 - n_2)$

(2) $(n_2 - n_1)$

(3) $\left(\frac{n_2 - n_1}{2}\right)$

(4) $(n_2 + n_1)$

22) A compound contains atoms A, B and C. The oxidation number of A is +1, B is + 2 and C is –1. The possible formula of the compound is :-

(1) ABC_2

(2) A_2BC

(3) ABC_3

(4) A_3BC

23) 6×10^{-3} mole $K_2Cr_2O_7$ reacts completely with 9×10^{-3} mole X^{+n} to give XO_3^- and Cr^{+3}. The value of n is :

(1) 1

(2) 2

(3) 3

(4) None of these

24) In which of the following reactions, H_2O_2 is acting as a reducing agent ?

(1) $Cl_2O_7(g) + H_2O_2(aq) \rightarrow ClO_2^-(aq) + O_2(g) + H^+$

(2) $H_2O_2(aq) + Fe^{+2}(aq) \rightarrow Fe^{+3}(aq) + H_2O(\square)$

(3) $N_2H_4(\square) + 2H_2O_2(\square) \rightarrow N_2(g) + 4H_2O(\square)$

(4) $H_2O_2 + SO_2 \rightarrow H_2SO_4$

25) Which of the following has been arranged in order of increasing oxidation number of nitrogen :-

(1) $NH_3 < N_2O_5 < NO < N_2$

(2) $NH_4^+ < NO_3^- < NO_2^- < N^{3-}$

(3) $NH_4^+ < N_2H_4 < NH_2OH < N_2O$

(4) $NO_2 < NaN_3 < NH_4^+ < N_2O$

26) Match the column

	Column-I		**Column-II**
(A)	Galvanic cell	(P)	Used in space programme
(B)	Electrolytic cell	(Q)	No transformation of electrical energy into chemical energy

(C)	Fuel cell	(R)	Cell reaction is spontaneous
(D)	Dead battery	(S)	Cell reaction is non spontaneous

Correct match is :

(1) A→S; B→Q; C→R; D→P,S

(2) A→R; B→S; C→P,R; D→Q

(3) A→S; B→R; C→P,Q; D→Q

(4) A→ P; B→S; C→Q,R; D→R

27) Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R :**

Assertion A : NaCl and KNO_3 have same value of constant 'A' in Debye-Huckel-Onsager equation.

Reason R : Both NaCl and KNO_3 are strong electrolytes.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

28) If $E^{o}_{Au^{+}/Au} = x$, $E^{o}_{Au^{3+}/Au} = y$ and $E^{o}_{Au^{3+}/Au^{+}} = z$ then correct relation will be :

(1) $3y = 2z + x$

(2) $y = x + z$

(3) $3y = 2z - x$

(4) $z = x + y$

29) The resistance of a conductivity cell filled with 0.1 M KCl solution is 100 ohm. If resistance of the same cell when filled with 0.02 M KCl solution is 250 ohm, calaculate the molar conductivity of 0.02 M KCl solution. The conductivity of 0.1 M KCl solution is 1.29 S m^{-1}.

(1) 2.58×10^{-2} S m^2mol^{-1}

(2) 124 S m^2mol^{-1}

(3) 1.153×10^{-3} S cm^2mol^{-1}

(4) 115×10^{-3} S m^2mol^{-1}

30) A certain current liberates 0.500 g of H_2 in 2 hours. How many gram of oxygen can be liberated by the same current in the same time?

(1) 0.500 g

(2) 8.00 g

(3) 4.00 g

(4) 16.00 g

31)

At 298 K, the standard free energy of formation of H_2O (□) is – 237.2 kJ/mol while that of its ionisation into H^+ ion and OH^- ion is 80 kJ/mol then emf of the following cell at 298 K will be.
H_2 (g, 1 bar) | H^+ (1M) || OH^- (1M) | O_2 (g, 1 bar)
[1F = 96500 C]

(1) –0.40 V

(2) 0.81 V

(3) 1.4 V

(4) 0.4 V

32) Given, $E^\circ_{Ag^+/Ag}$ = 0.80 V,
$E^\circ_{Mg^{2+}/Mg}$ = –2.37 V, $E^\circ_{Cu^{2+}/Cu}$ = 0.34 V,
$E^\circ_{Hg^{2+}/Hg}$ = 0.79 V.
Which of the following statements is correct

(1) $AgNO_3$ can be stored in copper vessel.

(2) $Cu(NO_3)_2$ can be stored in magnesium vessel.

(3) $CuCl_2$ can be stored in silver vessel.

(4) $HgCl_2$ can be stored in copper vessel.

33) Select the correct cell reaction of the cell
Pt (s) | Cl_2 (g) | Cl^- (aq) | | Ag^+ (aq) | Ag (s) :-

(1) $Cl_2(g) + Ag^+(aq) \rightarrow Ag(s) + 2Cl^-(aq)$

(2) $Cl_2(g) + Ag(s) \rightarrow 2Cl^-(aq) + Ag^+(aq)$

(3) $2Cl^-(aq) + 2Ag^+(aq) \rightarrow 2Ag(s) + Cl_2(g)$

(4) $AgCl(s) \rightarrow Ag^+(aq) + Cl^-(aq)$

34) Select the incorrect statement among the following.

(1) For SHE, SOP = 0 at all temperatures.

(2) E_{cell} is an intensive property.

(3) A positive E° means that the redox couple is stronger reducing agent than H^+/H_2 couple

(4) Magnitude of conductivity depends on the nature of the material

35) Find the charge in coulombs required to convert 0.2 mole VO_3^{2-} into VO_4^{3-}.

(1) 1.93×10^4

(2) 9.65×10^4

(3) 1.93×10^5

(4) 9.65×10^5

36)

Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column

II and assign the correct code :

	Column I		**Column II**
a.	Co^{3+}	i	$\sqrt{8}$ B.M.
b.	Cr^{3+}	ii	$\sqrt{35}$ B.M.
c.	Fe^{3+}	iii	$\sqrt{3}$ B.M.
d.	Ni^{2+}	iv	$\sqrt{24}$ B.M.
		v	$\sqrt{15}$B.M.

	a	b	c	d
(1)	iv	v	ii	i
(2)	i	ii	iii	iv
(3)	iv	i	ii	iii
(4)	iii	v	i	ii

(1) 1

(2) 2

(3) 3

(4) 4

37) Which of the following is incorrect about interstitial compound.

(1) they have high melting points in comparison to pure metals.

(2) they are very hard

(3) they retain metallic conductivity

(4) they are chemically very reactive

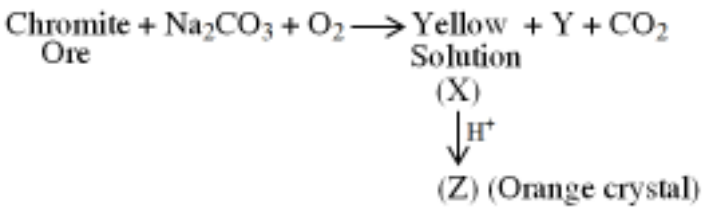
38) When $KMnO_4$ solution is added to oxalic acid, solution, the decolourisation is slow in beginning but becomes fast after some time because :-

(1) CO_2 is a by product

(2) Mn^{2+} acts as a auto catalyst

(3) reaction in exothermic

(4) MnO_4^- catalyzes the reaction



39)

Identify X, Y and Z respectively

(1) $Na_2Cr_2O_7$, Na_2CrO_4, Fe_2O_3

(2) Na_2CrO_4, $Na_2Cr_2O_7$, FeO

(3) Na_2CrO_4, Fe_2O_3, $Na_2Cr_2O_7$

(4) Na_2CrO_4, FeO, $Na_2Cr_2O_7$

40) Which of the following lanthanoids ion does not show colour.

(1) La^{+3}, Ce^{+3}

(2) Ce^{+3}, Lu^{+3}

(3) La^{+3}, Lu^{+3}

(4) La^{+3}, Ce^{+4}

41)

Most stable phosphorous is

(1) Red

(2) Yellow

(3) Black

(4) White

42) **Statement-I :** The value of enthalpy of atomisation is generally maximum at about the middle of each series.
Statement-II : There is one unpaired electron per d-orbital and this results in stronger interatomic interaction.

(1) Statement-I is True, Statement-II is True, Statement-II is a correct explanation for Statement-I.

(2) Statement-I is True, Statement-II is True, Statement-II is not a correct explanation for Statement-I.

(3) Statement-I is True, Statement-II is False.

(4) Statement-I is False, Statement-II is True.

43) Correct increasing order of number of π bonds in the following is:-
I. $H_2S_2O_6$ II. H_2SO_3 III. H_2SO_4

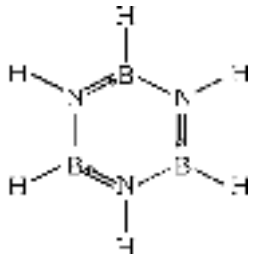
(1) I, II, III

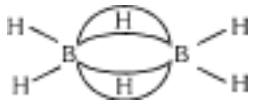
(2) II, III, I

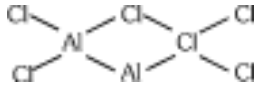
(3) II, I, III

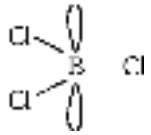
(4) I, III, II

44) Which of the following compounds is **not** matched correctly with its structure ?

(1)  – Borazine

(2)  – Diborane

(3)  – $AlCl_3$ (Dimer)

(4)  – Boron trichloride

45)

UF_6, an enrichment product of U^{235} is obtained as a product of reaction of Uranium with

(1) IF_7

(2) SF_4

(3) $XeOF_4$

(4) ClF_3

BIOLOGY

1) **Assertion (A) :** The genotype with all the dominant alleles (AABBCC) has the darkest skin colour and that with all the recessive alleles (aabbcc) has the lightest skin colour.
Reason (R) : In a polygenic quantitative inheritance, the phenotype reflects the contribution of each allele.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

2) **Assertion :** Independent gene pairs segregate independently of each other
Reason : The two chromosome pairs can align at the metaphase-I independently of each other.

(1) Both Assertion and Reason are true, but the 'Reason' is not a correct explanation of 'Assertion'.

(2) Assertion is true, but Reason is false.

(3) Both Assertion and Reason, are true and the 'Reason' is a correct explanation of 'Assertion'.

(4) Both the Assertion and Reason are false.

3) **Statement-I :-** Dominance is an autonomous feature of a gene.
Statement-II :- Dominance depends not only on gene product but also on the type of phenotype

that we choose to examine.

(1) Both statement-I and II are correct.
(2) Only statement-I is correct.
(3) Only statement-II is correct.
(4) Both statement-I and II are incorrect.

4) **Statement I :** Law of dominance is used to explain the expression of only one of parental characters in F_1 generation of monohybrid cross.
Statement II : Mendel selected 14 true breeding pea plant varieties in his hybridization experiments.

(1) Both statements I and II are correct.
(2) Both statements I and II are incorrect.
(3) Statement I is correct and statement II is incorrect .
(4) Statement I is incorrect and statement II is correct.

5) Match list-I with list-II :

	List-I		**List-II**
A.	$I^A i$	I	O blood group
B.	$I^A I^B$	II	A blood group
C.	i i	III	B blood group
D.	$I^B I^B$	IV	AB blood group

Choose the correct answer from options given below :-
(1) A-II, B-IV, C-III, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-I, D-II

6) Which one of the following options gives the correct categorisation of character according to the category A, B and C ?

	A (Character)	**B (Recessive trait)**	**C (Chromosome no. on which gene is present)**
(1)	Colour of cotyledon	Yellow	1
(2)	Colour of pod	Yellow	4
(3)	Colour of flower	White	1
(4)	Shape of seed	Round	7

(1) 1

(2) 2

(3) 3

(4) 4

7) A diploid organism is heterozygous for 3 loci & homozygous for 2 loci. How many types of gametes will be produced by it?

(1) 128

(2) 8

(3) 4

(4) 14

8) Which one of the following characters chosen by Mendel can express in both homozygous as well as in heterozygous condition?

(1) Axial flower position

(2) Constricted pod shape

(3) Wrinkled seed shape

(4) Green seed colour

9) **Statement-I :** Segregation of paired factor takes place at the time of anaphase-I of meiosis.
Statement-II : Independent assortment occurs when genes are located on different pairs of chromosome.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

10) A heterozygous round seeded pea plant is crossed with wrinkled seeded pea plant. How many plants will produce intermediate sized starch grains in their seeds, out of 2000 progenies obtained?

(1) 0

(2) 1500

(3) 1000

(4) 750

11) In corn, the trait for tall plant (T) is dominant to the trait for dwarf plant (t) and the trait for coloured kernels (C) is dominant to the trait for white kernels (c). In a particular cross of corn plant, the probability of an offspring being tall is 0.5 and probability of a kernel being coloured is 0.75. Which of the following most probabaly represents the parental genotypes?

(1) TtCc × TtCc

(2) TtCc × ttCc

(3) TtCc × ttcc

(4) TTCc × ttCc

12) The minimum weight of tomatoes obtained from a plant is 20g. The maximum weight of tomatoes obtained is 120g. What will be the weight of tomatoes obtained by crossing plants with genotype AAbb × aabb ?

(1) 45g
(2) 70g
(3) 100g
(4) 25g

13) How many characters of pod of pea plant studied by Mendel during his experiments ?

(1) 4
(2) 2
(3) 7
(4) 1

14) What will be the percentage of heterozygous plant in F_3 generation of Mendel's monohybrid cross ?

(1) 25%
(2) 50%
(3) 75%
(4) 12.5%

15) In a garden pea plant, a dihybrid cross was performed. In F_2 generation 9/16 members were showing same phenotype. The phenotype was :

(1) Violet and terminal flower
(2) Violet flower and green pod
(3) Green and round seed
(4) Tall stem and white flower

16) In a trihybrid plant, what will be the probability of gamete which contains at least one dominant allele ?

(1) $\frac{1}{8}$
(2) $\frac{3}{8}$
(3) $\frac{7}{8}$
(4) $\frac{1}{2}$

17) Who observed that behaviour of chromosomes at meiosis can serve as the cellular basis of both segregation and independent assortment of Mendelian factors ?

(1) Sutton and Boveri

(2) Banden and Boveri

(3) W. Fleming

(4) Boveri and Braurer

18) In case of polygenic quantitative inheritance, if a trait is controlled by three genes, then in F_2 generation what will be the number of dominant alleles in the organism which is obtained in maximum number ?

(1) 6

(2) 4

(3) 3

(4) 2

19) Occasionally, a single gene may express more than one effects. This is called :

(1) Polygenic inheritance

(2) Pleiotropy

(3) Multiple allelism

(4) Co-dominance

20) The types of genotypes produced, when individuals of genotype YyRrTt are crossed with each other, are :

(1) 4

(2) 45

(3) 28

(4) 27

21) Find out the frequency of plants which are homozygous for both genes in a dihybrid test cross.

(1) $\frac{1}{2}$

(2) $\frac{3}{4}$

(3) $\frac{1}{4}$

(4) $\frac{1}{16}$

22) Which of following blood groups is possible only in homozygous condition ?

(1) O^+

(2) AB^-

(3) B^-

(4) O^-

23) The unmodified allele which represents the original phenotype is the __(a)__ and the modified allele is __(b)__ the __(c)__ allele :
Fill in the blanks with suitable option

(1) (a) Recessive (b) generally (c) dominant

(2) (a) Dominant (b) always (c) recessive

(3) (a) Dominant (b) generally (c) recessive

(4) (a) Recessive (b) always (c) dominant

24) How many hormones from the below list interact with intracellular receptors and regulate chromosome function and gene expression.
GH, Oxytocin, ADH, ANF, Aldosterone, Cortisol, Thyroxin, Epinephrine, Estradiol, GnRH, Progesterone

(1) Four

(2) Five

(3) Six

(4) Seven

25)

Read the following **statements** (A–D) :

(A) In the resting phase of nerve sodium ion channel are closed and potential is found +70mv at inside (B) +10 mv potential difference from resting state generate threshold stimulus
(C) Na^+ increases more in ECF in depolarisation which cause generation of action potential
(D) The time taken for restoration of resting state is called refractory period
How many of above statements are incorrect?

(1) 4

(2) 3

(3) 2

(4) 1

26) Read the following four statements (A - D) :
(A) Both, repolarisation and hyperpolarisation are activity of K^+ VGC closing
(B) Schwan cell found in both myelinated and non-myelinated neurons of PNS
(C) Below skull duramater, piamater and arachnoid meninges are located in order
(D) During resting stage ionic gradient across membrane is maintained by activity of $Na^{\oplus}$ – $K^{\oplus}$ pump.
How many of the above statements are right ?

(1) Four

(2) One

(3) Two

(4) Three

27) Consider the following four statement (A–D) and select option which include all the correct ones only:-

(A) Acromegaly is due to hyper secretion of STH in childhood stage.
(B) In adulthood stage hypothyroidism cause ‘‘Gull's disease’’.
(C) Hyper secretion of glucocorticoids cause "Addison's disease".
(D) At the time of physical labour or fasting taking of insulin injection can cause ‘insulin shock’ in diabetic patient.

(1) Statements B & C
(2) Statements B & D
(3) Statements A & B
(4) Statements A & C

28) How many statements are not **correct** ?
(i) FSH stimulates ovulation in females. As a result corpus albicans is formed.
(ii) Secretion of thyroxine is stimulated by thyrotropin releasing factor (TRF) of anterior pituitary.
(iii) Prolactin is responsible for formation of milk.
(iv) GH increases uptake of glucose in the cells, so it is also called diabetogenic hormone.

(1) 1
(2) 2
(3) 3
(4) 4

29) Read the following statements (A-D) :-
(A) The nerve fibres of PNS are two types afferent and efferent fibres
(B) PNS comprises of all nerves of body associated with the brain only
(C) Efferent fibres transmit regulatory impulses from CNS to peripheral tissues
(D) Brain, ganglia and neural tissue found in neural system of insecta.
How many of above statement are right ?

(1) Two
(2) Three
(3) One
(4) Four

30) Select the option with incorrect combination of the structure with its location and function :-

	Structure	**Location**	**Function**
(1)	Corpora quadrigemina	Mid brain	Visual and auditory reflexes
(2)	Thalamus	Wrapped by cerebrum	Major coordinating centre for sensory & motor signalling
(3)	Limbic system	Inner part of cerebral hemisphere	Emotions, sexual behavior and motivation
(4)	Motor area	Frontal lobe	Sensory stimulus perception

(1) 1

(2) 2

(3) 3

(4) 4

31) **Statement - I :-** The potential difference across the membrane at rest is called the resting membrane potential and this is about - 70 m/s.
Statement - II :- Axolemma is selectively permeable in nature.
Choose the correct option from the given below -

(1) Both statement - I and statement - II are incorrect.

(2) Statement - I is correct but statement - II is incorrect.

(3) Statement - I is incorrect but statement - II is correct.

(4) Both statement - I and stateeent - II are correct.

32) Find out the correct matches :-

Column-I		Column-II	
(A)	Broca's area	(i)	Temporal lobe
(B)	Association area	(ii)	Hypothalamus
(C)	Aqueduct	(iii)	Pons
(D)	Olfactory area	(iv)	Cerebral cortex
(E)	Relay centre	(v)	Speech area
(F)	Body temprature regulation	(vi)	Mid brain
		(vii)	Thalamus

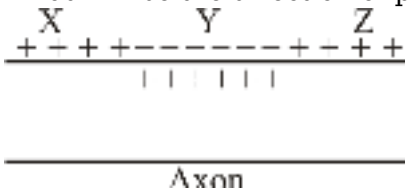
(1) A-iv, B-i, C-vi, D-vii, E-iii, F-ii

(2) A-v, B-vi, C-iv, D-i, E-ii, F-iii

(3) A-vi, B-vii, C-i, D-iii, E-ii, F-iv

(4) A-v, B-iv, C-vi, D-i, E-vii, F-ii

33) The given figure illustrates the axonal membrane and propagation of nerve impulse.
What will be the direction of propagation of the nerve impulse out side the axon?



(1) X → Y

(2) Y → X

(3) Z → Y

(4) Z → X

34) Match Column-I with Column-II :-

	Disease		**Deficient Neurotransmitter**
(A)	Parkinson's disease	(i)	Acetylcholine
(B)	Huntington's chorea	(ii)	Dopamine
(C)	Alzheimer's disease	(iii)	GABA

(1) A - i, B - ii, C - iii

(2) A - ii, B - iii, C - i

(3) A - iii, B - ii, C - i

(4) A - iii, B - i, C - ii

35) **Assertion :-** Iodine is essential for the normal rate of hormone synthesis in thyroid.
Reason :- Deficiency of iodine in our diet results in enlarge thyroid gland known as Exophthalmic goitre.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

36) ***Assertion (A) :-*** Cortisol produces anti-inflamatory reactions and suppress the immune response.
Reason (R) :- Cortisol stimulates gluconeogenesis, lipogenesis and proteolysis.

(1) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(3) **(A)** is correct but **(R)** is not correct

(4) **(A)** & **(R)** both incorrect

37) **Assertion :** Catecholamines are rapidly secreted in response to stress of any kind during emergency situation.
Reason : Catecholamines are called emergency hormone or hormones of fight or flight.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

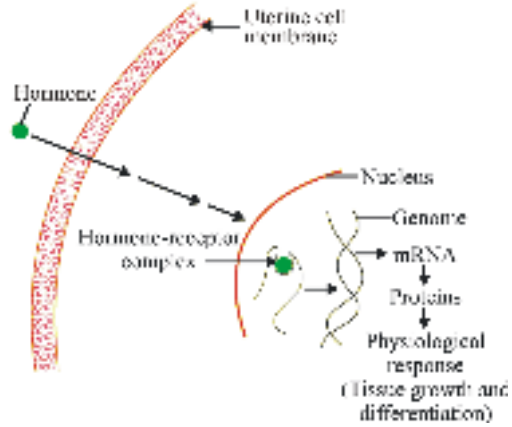
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

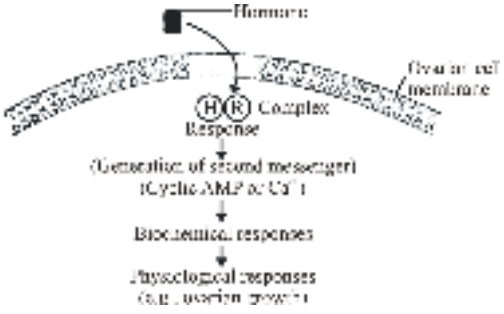
(4) Both assertion & reason are false.

38) In following diagrame which shows the working of Proteins hormone ideally :-

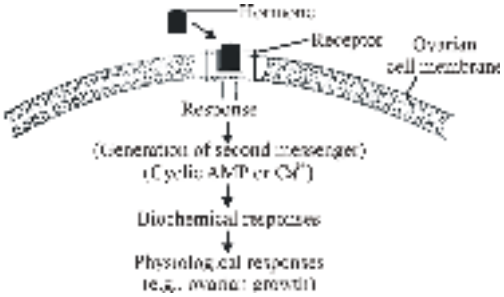
(1)



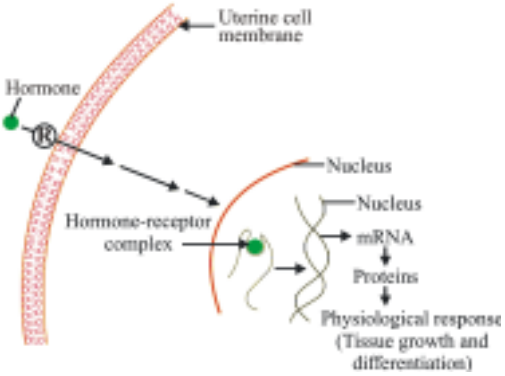
(2)



(3)



(4)



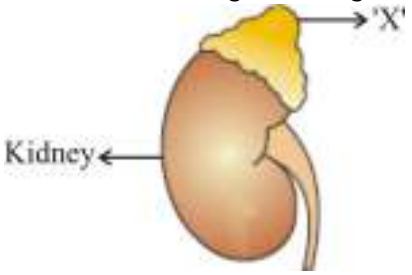
(1) 1

(2) 2

(3) 3

(4) 4

39) Observe the given diagram :-



(a) Adrenal gland.
(b) Suprarenal gland.
(c) Gland which secrete mineralo-corticoid.
(d) Responsible for Addison's disease.
How many of the above represent 'X' ?

(1) 3
(2) 1
(3) 4
(4) 2

40) Which of the following options gives the correct group of hormones related to their functions :-

(i)	Thyroxine	A = Calorigenic hormone
		B = Nodule formation during hyposecretion
(ii)	Melatonine	A = Antigonadial activity
		B = Helps in myelinogenesis in human
(iii)	Thymosin	A = Cellular immunity by T-Lymphocyte
		B = Cushing disease
(iv)	ADH	A = Stimulates mainly Ionic reabsorption
		B = Hyposecretion causes Diabetes insipidus

(1) (i)-B (ii)-B (iii)-A (iv)-A
(2) (i)-A (ii)-A (iii)-A (iv)-B
(3) (i)-A (ii)-B (iii)-A (iv)-B
(4) (i)-B (ii)-B (iii)-A (iv)-B

41) Addison's disease is caused by :-

(1) Hyposecretion of thyroid gland
(2) Hyposecretion of adrenal gland
(3) Hypersecreton of pineal gland
(4) Hyposecretion of pituitary gland

42) A pregnant female delivers a baby who suffers from stunted growth, mental retardation, low intelligence quotient and abnormal skin. This is the result of:

(1) Over secretion of pars distalis
(2) Deficiency of iodine in diet
(3) Low secretion of growth hormone
(4) Cancer of the thyroid gland

43) Due to malfunctioning of insulin hormone, what effect would be on blood sugar and disease occur :

(1) Hyperglycemia, Diabetes Mellitus
(2) Hypoglycemia, Diabetes Insipidus
(3) Hypoglycemia, Diabetes Mellitus

(4) Hyperglycemia, Diabetes Insipidus

44) From the followings how many hormones are protein in nature :-
Cortisol, Hypothalamic hormone, Pituitary hormone, Testosterone, Estradiol, Progesterone, Epinephrine.

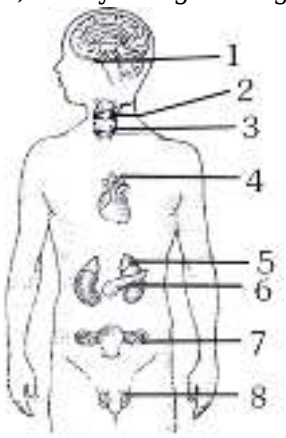
(1) 4

(2) 5

(3) 6

(4) 3

45) Study the given figure and choose the correct option -



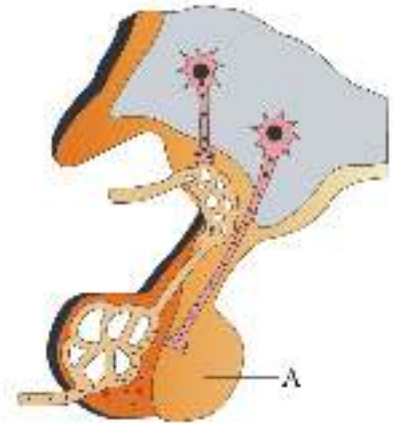
(1)
2. → Parathormone
6. → islets of langerhans
8. → ICSH production
3. → Thyroxine

(2)
5. → Adrenaline
1. → FSH production
8. → Testosterone
4. → Immunity

(3)
1. → Master gland
8. → ICSH production
2. → Thyroxine
5. → Adrenaline

(4)
7. → LH production
8. → Testosterone
6. → Insulin
4. → Thyroxine

46) Which of the following diseases are related to the hormone secreted from the structure marked 'A' in the given figure :-



A) Goitre B) Cretinism
C) Acromegaly D) Gigantism
E) Addision disease F) Diabetes insipidus

(1) A, B, C, D, E
(2) C, D, E
(3) F only
(4) C only

47) Match the plant with the kind of life cycle it exhibits :-

	List-I		**List-II**
(a)	*Volvox*	(i)	Diplohaplontic
(b)	*Ectocarpus*	(ii)	Diplontic
(c)	*Fucus*	(iii)	Haplontic
(d)	*Fern*	(iv)	Haplodiplontic

(1) a-ii, b-iv, c-ii, d-i
(2) a-iii, b-iv, c-i, d-ii
(3) a-iv, b-iii, c-ii, d-i
(4) a-i, b-ii, c-iii, d-iv

48)

The first cell of sporophytic generation in bryophyta is:-

(1) Spore
(2) Spore mother cell
(3) Zygote
(4) Protonema

49)

The unique feature of Bryophytes compared to other green plant groups is that:-

(1) They produce spores

(2) They lack vascular tissue
(3) They lack root
(4) Their sporophyte is attached to gametophyte

50) Which of the following is not correctly matched?

(1) Lycopsida – *Selaginella*
(2) Psilopsida – *Lycopodium*
(3) Sphenopsida – *Equisetum*
(4) Pteropsida – *Adiantum*

51) Most reduces male gametophyte and sporophyte are present respectively :-

(1) *Marchantia* and *lilium*
(2) *Wolfia* and *volvox*
(3) *Funaria* and *fern*
(4) *Dryopteris* and *Polysiphonia*

52) Choose the **incorrect** statement from followings.

(1) The most advanced gametophyte is found in mosses.
(2) Highly reduced gametophyte, having limited numbers of cells is found in flowering plants.
(3) In both bryophytes and pteridophytes sporophyte is free living.
(4) Seeded plants show diplontic life cycle

53) What is correct :-

(1) Protonema of moss and prothallus of Pteridium are sporophytic.
(2) Protonema of moss and prothallus of *Pteridium* are gametophytic
(3) Moss protonema is sporophytic, *Pteris* prothallus is gametophytic but plant body of *Pteridium* and moss are gametophytic.
(4) Plant body of moss is gametophytic while that of *Pteridum* is both gametophytic and sporophytic

54)

Independent free living, photosynthetic gametophyte is not found in :

(1) *Funaria*
(2) *Marchantia*
(3) *Cycas*
(4) *Riccia*

55)

Which of the following plants group do not produced seeds?

(1) *Pinus* and *Funaria*

(2) *Chlamydomonas* and *Pinus*

(3) *Pisum* and *Pinus*

(4) *Funaria* and *Chara*

56) Identify the following figures labelled as A, B, C and D.

(A)  (B) 

(C)  (D) 

	A	B	C	D
(1)	*Pinus*	*Chlamydomonas*	*Salvinia*	*Funaria*
(2)	*Ginkgo*	*Fucus*	*Equisetum*	*Sphagnum*
(3)	*Ginkgo*	*Volvox*	*Salvinia*	*Sphagnum*
(4)	*Cycas*	*Volvox*	*Sphagnum*	*Selaginella*

(1) 1

(2) 2

(3) 3

(4) 4

57) 

Identify the A, B, C and D

(1) A - *Equisetum*, B - *Paramoecium*,
C - *Funaria*, D - *Sphagnum*

(2) A -*Euglena*, B -*Agaricus*,
C -*Ginkgo*, D-*Nostoc*

(3) A - *Agaricus*, B - *Ginkgo*,
C - *Euglena*, D - *Equisetum*

(4) A - *Nostoc*, B - *Euglena*,
C - *Agaricus*, D - *Ginkgo*

(1) 1

(2) 2

(3) 3

(4) 4

58) Identify the given figures of algae and select the correct option.

(A)  (B) 

(C)  (D) 

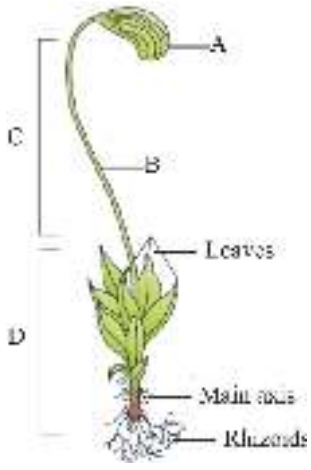
(1) A–*Volvox*, B–*Chlamydomonas*, C–*Chara*, D–*Porphyra*

(2) A–*Fucus*, B–*Dictyota*, C–*Porphyra*, D–*Polysiphonia*

(3) A–*Fucus*, B–*Polysiphonia*, C–*Porphyra*, D–*Dictyota*

(4) A–*Dictyota*, B–*Porphyra*, C–*Fucus*, D–*Polysiphonia*

59) In given figure A,B,C and D represent : -



(1) A-Capsule, B-seta, C-Gametophyte, D-Sporophyte

(2) A-Capsule, B-Seta, C-Sporophyte, D-Gametophyte

(3) A-foot, B-Seta, C-Gametophyte, D-Sporophyte

(4) A-foot, B-Capsule, C-Gametophyte, D-Sporophyte

60) Identify the true statements for the plant shown in figure and select the correct option given



below :-

(A) Monoecious gametophyte of this plant require cool, damp, shady place to grow

(B) Antheridia and archegonia both are formed on same gametophyte

(C) It produce two different type of spore.

(D) The development of the zygote into young embryo takes place within the female gametophyte.

(1) A, B, C, & D

(2) B, C & D only

(3) C & D only

(4) A, C & D only

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	2	1	4	2	3	4	2	3	4	4	3	3	1	2	4	3	1	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	4	1	3	2	3	2	4	3	2	3	4	4	3	3	4	3	2	1
Q.	41	42	43	44	45															
A.	4	3	2	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	3	1	2	2	3	4	3	2	2	4	1	2	1	1	4	1	4	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	3	1	1	3	2	2	1	1	3	4	3	3	3	1	1	4	2	3	3
Q.	86	87	88	89	90															
A.	3	1	2	3	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	3	1	2	3	2	1	1	3	2	1	2	1	2	3	1	3	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	3	2	3	3	2	3	2	4	4	4	3	2	2	3	1	3	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	1	4	2	3	1	3	4	2	2	3	2	3	4	3	4	3	2	3

SOLUTIONS

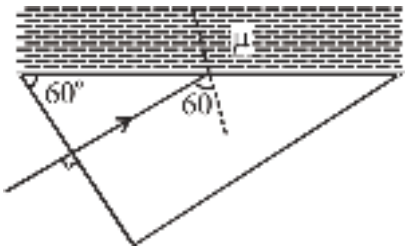
PHYSICS

1)

Same f^0 and same MP $\left(\frac{f_0}{f_e}\right)$ ⇒ same f_e

$\therefore f_{e,\,GT} = f_{e,AT} = f$

$$\left.\begin{array}{l}\text{Length of tube of GT} = f_0 - f \\ \text{Length of tube of AT} = f_0 + f\end{array}\right] \text{ difference} = 2f$$



2)

$$\sin 60 \geqslant \frac{\mu}{3/2}$$

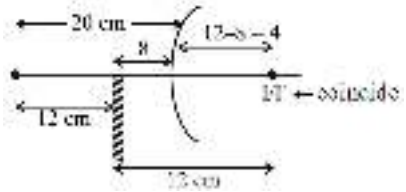
$$\sqrt{\frac{3}{2}} \times \frac{3}{2} \geqslant \mu$$

3) f = –5cm, u = –20cm

$$\frac{h_i}{L_i} = \frac{m_T}{m_L} = \frac{(-v/u)}{-(v/u)^2} \Rightarrow \frac{u}{v} = \frac{u-f}{f} = \frac{(-20)-(-5)}{-5}$$

$$= \frac{-15}{-5} = \frac{3}{1}$$

4) Image by plane mirror form behind 12 cm from plane mirror



V = +4 or U = – 20

$$\frac{1}{F} = \frac{1}{V} + \frac{1}{U}$$

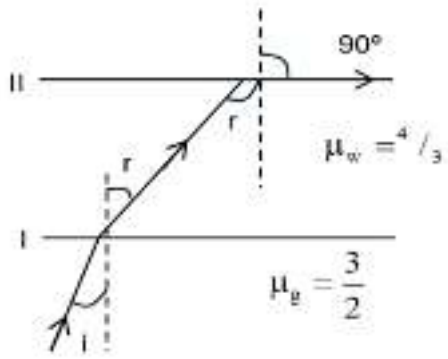
$$\frac{1}{F} = \frac{1}{4} - \frac{1}{20}$$

F = 5 cm

5)

r is the critical angle

$$\sin r = \frac{1}{\mu_w} = \frac{3}{4}$$



For surface I,

$\sin i \times \mu_g = \sin r \times \mu_w$

$\sin i \times \frac{3}{2} = \frac{3}{4} \times \frac{4}{3}$

$\sin i \times \frac{3}{2} = 1$

Statement 2 – Correct and Statement 1 - False

6)

$f_0 = 1.2$ cm, $f_e = 3$ cm

$u_0 = -1.25$ cm

$\frac{1}{v_0} - \frac{1}{-1.25} = \frac{1}{1.2} \Rightarrow \frac{1}{v_0} = \frac{1}{1.2} - \frac{1}{1.25}$

$v_0 = 30$ cm

$m^0 = \frac{v_0}{u_0} = \frac{30}{-1.25} = -24$

final image is formed at infinity

So, $m_e = \frac{d}{f_e} = \frac{25}{3}$

magnifying ower of microscoe

$M = m^0 m_e = -24 \times \frac{25}{3} = -200$

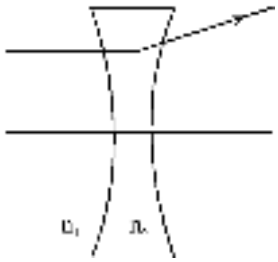
7) By applying lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} = \frac{f+u}{uf}$; $m = \frac{v}{u} = \frac{f}{f+u}$; $\frac{1}{m} = \frac{u}{f} + 1$

Slope of $\frac{1}{m}$ and 'u' is $\frac{1}{f}$

$\therefore \frac{1}{f} = \frac{b}{c}$ so $f = \frac{c}{b}$

8)



$n_2 > n_1$

9) d : distance between lenses

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

$$P = \frac{1}{F} = 0$$

$$\frac{d}{f_1 f_2} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{f_2 + f_1}{f_1 f_2}$$

$$d = f_1 + f_2$$

10)

Focal length of lens in a medium.

$$f_m = \left[\frac{\mu - 1}{\frac{\mu}{\mu_m} - 1}\right] f_a$$

Given $\mu = \mu$, $\mu_m = \mu_0$

Power in air = $P_a = \frac{1}{f_a}$

$$P_m = \frac{1}{f_m}$$

$$= \frac{1}{f_m} = \left[\frac{\frac{\mu}{\mu_m} - 1}{\mu - 1}\right] P_a \Rightarrow P_m = \left[\frac{\mu - \mu_0}{\mu - 1}\right] \frac{P}{\mu_0}$$

11)

$$f_m = f_a \left(\frac{\mu_\ell - 1}{\mu_\ell - \mu_m}\right) \times \mu_m$$

Since $\mu_m > \mu_{\square}$, therefore f_m will be opposite as $(\mu_{\square} - \mu_m) < 1$.

12) If the central part of the aperture up to diameter (d/2) is blocked by an opaque paper, then the focal length of the lens will not change.

 Intensity α (aperture)^2

$$\frac{I_1}{I_2} = \frac{d_1^2}{d_2^2} \Rightarrow \frac{I}{I_2} = \frac{d^2}{d^2 - \left(\frac{d}{2}\right)^2} = \frac{1}{3/4}$$

$\Rightarrow I_2 = \left(\frac{3I}{4}\right)$

13) $\mu = \frac{\lambda_a}{\lambda_m} = \frac{3}{2}$

$\frac{1}{f} = \frac{\mu - 1}{R} = \frac{1}{2R}$

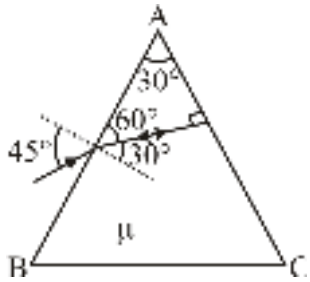
$\Rightarrow \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\Rightarrow \frac{1}{8} - \frac{1}{-24} = \frac{1}{2R}$

$\Rightarrow \frac{3+1}{24} = \frac{1}{2R}$

$\Rightarrow$ R = 3m.

14)



By snell's law

$1 \times \sin 45° = \mu \sin 30°$

$\frac{1}{\sqrt{2}} = \mu\left(\frac{1}{2}\right)$

$\Rightarrow \mu = \sqrt{2}$

15)

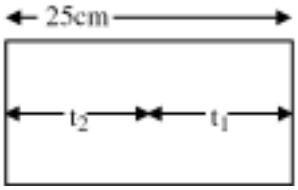
$V_{BF} = \mu v_B + V_F$

$9 = \frac{4}{3} v_B + 3$

$6 = \frac{4}{3} v_B \Rightarrow v_B = \frac{18}{4}$

$\Rightarrow v_B = 4.5$ m/s

16)



$\frac{t_1}{\mu} = 6$, $\frac{t_2}{\mu} = 4$

$t_1 = 6\mu$, $t_2 = 4\mu$

$t_{net} = t_1 + t_2 = 25$

$= 6\mu + 4\mu = 25$

$10\mu = 25$
$\mu = 2.5$

17) Lateral shift = $\frac{t\sin(i-r)}{\cos r}$
Here $1.\sin 60° = \sqrt{3}\sin r$
⇒ $r = 30°$, $t = 5\sqrt{3}$cm
Lateral shift = $5\sqrt{3}\frac{\sin(60°-30°)}{\cos 30°} = 5$cm

18)

$V_1 = -m^2 \times V_0$
$m = \frac{f}{f-u} = \frac{-24}{-24-(-60)} = \frac{-24}{36} = \frac{-2}{3}$

$V_I = -\left(\frac{-2}{3}\right)^2 \times 9$ $V_I = -4$ cm/sec (away from the mirror)

19)

Flux through
S_1 (ϕ_1) = $\frac{\text{Charge enclosed within } S_1}{\varepsilon_0} = \frac{q}{\varepsilon_0}$
Since the shell is conducting therefore a charge of magnitude -q will be induced at the inner surface (radius R_1) of the shell.
 Flux through
S_2 (ϕ_2) = $\frac{\text{Charge enclosed within } S_2}{\varepsilon_0} = \frac{q-q}{\varepsilon_0} = 0$

20)

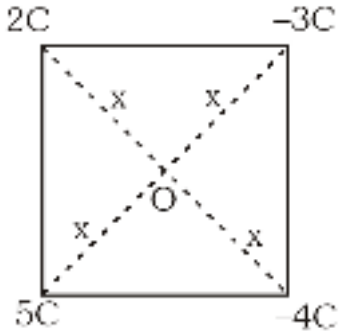
Time period
$T = 2\pi\sqrt{\frac{m}{2k}}$ or $\omega = \sqrt{\frac{2k}{m}}$
for amplitude
$2kA = \frac{\sigma q}{\varepsilon_0}$
or
$A = \frac{\sigma q}{2\varepsilon_0 k}$

21) $F = qE$
$F = q\frac{\sigma}{\varepsilon_0}$
if one plate is removed

F' = $q\frac{\sigma}{2\varepsilon_0} = \frac{F}{2}$

22)

$V_O = \frac{k(2C)}{x} + \frac{k(-3C)}{x} + \frac{k(-4C)}{x} + \frac{k(5C)}{x}$



$V_O = 0, E_O \neq 0$

23) E = 0 outside as $q_{enc} = 0$ (by gauss theorem)

24) $\frac{\partial V}{\partial x} = 6x = 6 \times 1 = 6$, $\frac{\partial v}{\partial y} = -2y = -2\times1 = -2$,
$\frac{\partial V}{\partial x} = 4z = 4\times1 = 4$

$\therefore \vec{E} = -\left[\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right] = -6\hat{i} + 2\hat{j} - 4\hat{k}$ V/m

$E = \sqrt{6^2 + 2^2 + 4^2} = \sqrt{56}$ V/m

25) (1) E.F outside a conducing charged sphere.

$E = \frac{KQ}{r^2} \Rightarrow E \propto \frac{1}{r^2}$

(2) Electric potential outside a charged & conducting sphere.

$V = \frac{KQ}{r} \Rightarrow V \propto \frac{1}{r}$

(3) E.F inside ∝ non conducting charged sphere.

26)

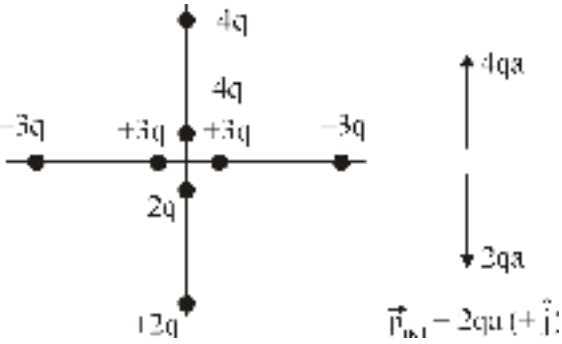
F = qE

$a = \frac{F}{m} = \frac{qE}{m}$

$m_p >> m_e$

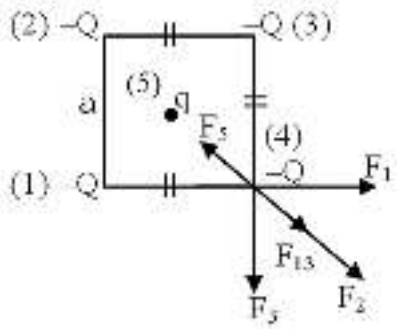
$|Q_p| = |Q_e| = e$

27)



$\vec{F}_{net} = 2qa\,(+\hat{j})$

28)



$[F_{net}]$ position – 4 = 0

$F_{13} + F_2 - F_5 = 0$

$$\sqrt{2}\frac{KQQ}{a^2} + \frac{KQQ}{\left(\sqrt{2a}\right)^2} - \frac{KQQ}{\left(\frac{a}{\sqrt{2}}\right)^2} = 0$$

$$\frac{KQ^2}{a^2}\left[\sqrt{2} + \frac{1}{2}\right] = \frac{KQq}{a^2} \times 2$$

$$Q\left[\sqrt{2} + \frac{1}{2}\right] = 2q$$

$$q = \frac{Q}{4}\left[1 + 2\sqrt{2}\right]$$

29)

$$T = \frac{2V_y}{g} = \frac{2 \times 10}{10} = 2 \text{ seconds}$$

$$H = \frac{V_y^2}{2g} = \frac{(10)^2}{2 \times 10} = 5\text{m}$$

$$R = \frac{1}{2}a_x T^2 = \frac{1}{2} \times \left(\frac{qE}{m}\right)(T^2)$$

$$R = \frac{1}{2} \times \left[\frac{10^{-3} \times 10^4}{2}\right](2)^2$$

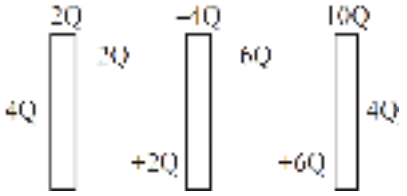
R = 10 metre.

30)

For smaller drop $v = \frac{Kq}{r}, v_{big} = \frac{KQ}{R}$

Here - q = 1000 q and $R = (1000)^{1/3}\, r = 10r$

$V_{big} = \dfrac{K(1000)Q}{10r} = 100\dfrac{KQ}{r} = 100V$

31) 

$Q_{net} = Q_{sys} = 2Q + (-4Q) + 10Q = 8Q$

32)

The direction of the field depends on the sign of the charge.
• For a positive charge, the electric field points away from the charge.
• For a negative charge, the electric field points towards the charge.
To ensure the electric field is negative at a point between Q_1 and Q_2, the direction of the fields due to both charges must add up in the negative direction of x. This can happen if Q_1 is negative and Q_2 is positive and the graph is symmetric so $|Q_1| = |Q_2|$

33)

Statement A ⇒ [True] Coulomb's law follow inverse square law.

Statement B ⇒ [True] E.F in the vicinity of charge conductor of irregular shape $\dfrac{\sigma}{\varepsilon_0}\hat{n}$
Statement C ⇒ [False] Because E.F. is depend on both charge inside or outside the closed surface.

34) $R_{eq} = \dfrac{2Rr}{R+r}$

$i_{Total} = \dfrac{E}{R_{eq}} = \dfrac{E(R+r)}{2Rr}$

$i_{AB} = \dfrac{r}{R+r} i_{Total}$

$i_{AC} = \dfrac{R}{R+r} i_{Total}$

$i_{BC} = i_{AC} - i_{AB}$ (from C to B)

$= \dfrac{R-r}{R+r} i_{Total}$

$= \dfrac{R-r}{R+r}\left(\dfrac{E(R+r)}{2Rr}\right) = \dfrac{E(R-r)}{2Rr}$

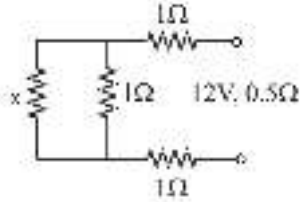
35)

$R = \dfrac{V}{I_g} - Rg = \dfrac{5}{100 \times 10^{-6}} - 100$

$= 5 \times 10^4 - 100 = 49900\ \Omega$

36) Let the effective resistance of the network is x. If one part of the network has resitstance (1Ω, 1Ω, 1Ω) is separated as shown, the effective resistance remains x (as it is infinitely network). Here, x and 1Ω are in parallel :

$$\frac{1}{R'} = \frac{1}{x} + \frac{1}{1} = \frac{1+x}{x} \Rightarrow R_P = \frac{x}{1+x}$$



Now, resistances R', 1Ω and 1Ω are in series. So, the resultant resistance :

$$R = R' + 1 + 1 = \frac{x}{1+x} + 1 + 1 = \frac{x}{1+x} + 2 \quad \text{.....(i)}$$

In case of infinite resistances, the value of R remains x

$$\therefore \quad x = \frac{x}{1+x} + 2$$

$\Rightarrow$ $x(x + 1) = x + 2 + 2x \Rightarrow x^2 - 2x - 2 = 0$

$$x = \frac{-(-2) \pm \sqrt{4+8}}{2} = \frac{2 \pm \sqrt{12}}{2} = 1 \pm \sqrt{3}$$

The value of resistance cannot be negative. So, the resistance of network x $= 1 + \sqrt{3} = 1 +$ 1.732

x = 2.732 Ω

Total resistance of the circuit = 2.732 + 0.5 = 3.232 Ω

Current drawn from the supply

$$I = \frac{V}{3.232} = \frac{12}{3.232} = 3.72 \text{ A}$$

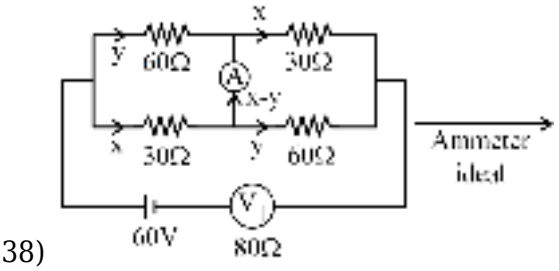
37)

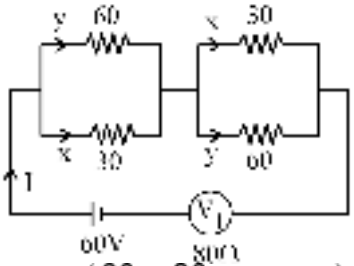
1A = 1000 mA

$900 \times 10^{-3} S = 100 \times 10^{-3} \times 1000$

$$\Rightarrow S = \frac{1000}{9} = 111\,\Omega$$

38)





$$R_{eq} = \left(\frac{60 \times 30}{60+30} \times 2 + 80\right) \Omega$$

= 120 Ω

Total current, $I = \frac{60}{120} = \frac{1}{2}A$

So, $x = \frac{60}{60+30} \times \frac{1}{2} = \frac{1}{3}A$

and $y = \frac{30}{60+30} \times \frac{1}{2} = \frac{1}{6}A$

Reading of ammeter = x – y = $\frac{1}{3} - \frac{1}{6} = \frac{1}{6}A$

Reading of V_1 = I (80) = $\frac{1}{2}$ (80) = 40 V

Reading of V_2 (ideal) = x(30) = $\frac{1}{3} \times 30 = 10$ V

39) Total voltage = $\frac{8volt}{km} \times 150km$ = 1200 Volt

Total resistance = 0.5 Ω/km × 150 km = 75 Ω

$P = \frac{V^2}{R} = \frac{(1200)^2}{75}$

$P = \frac{1440}{75} \times 10^3 = 19.2kW$

40)

Apply KVL from A to B

V_A – i(4) – 9 – (i – i_1) + 3 – 4i – V_B = 0

But V_A – V_B = 16V

10 – 9i + i_1 = 0 ⇒ 9i – i_1 = 10 ...(i)

and –9 – (i – i_1) + 2i_1 = 0 ⇒ –i + 3i_1 = 9 ....(ii)

9 × eq^n (ii) = –9i + 27i_1 = 81 ....(iii)

From (i) and (iii) 26i_1 = 91 ⇒ $i_1 = \frac{91}{26} = \frac{7}{2}$ = 3.5A

41)

Potential difference across PQ i.e. p.d. across the resistance of 20Ω, which is V = i × 20

and $i = \frac{48}{(100+100+80+20)} = 0.16A$

$\therefore V = 0.16 \times 20 = 3.2V$

42) $I = \frac{(q_{net})\omega}{2\pi} = \frac{(2Q)\omega}{2\pi}$

43) $\frac{V+1+1}{3} + \frac{V-5-2}{7.5} + \frac{V-6-6}{5} = 0$

V_0 = 4V

$$I_{4\Omega} = \frac{4+1+1}{3} = 2A$$

44)

This question is based on for maximum power transfer, external resistance R = r_{net}.
Given internal resistance in series : r_{net} = 2 + 1 = 3Ω.
Hence, for maximum power, R = r_{net} = 3Ω.

45) $\frac{R_1 + 10}{R_2} = \frac{50}{100-50} = 1$
⇒ $R_2 = R_1 + 10$ ...(1)
⇒ Now without 10 Ω, $\frac{R_1}{R_2} = \frac{40}{60}$
⇒ $R_2 = \frac{3}{2}R_1$ ...(2) by (1) & (2)
R_1 = 20 Ω

CHEMISTRY

46)

2 NO $\rightleftharpoons$ $N_2 + O_2$
ΔH < 0
$N_2 + O_2 \rightleftharpoons$ 2NO
ΔH > 0
for endothermic Reaction
if T ↑, K ↑

47) $K_P = K_C (RT)^{\Delta n_g}$
Δn_g = gaseous moles

48)

Concepts:

A. **Complex Formation:**

Analysis:

A. **Assertion (A):**

A. **The reaction $Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$ produces a red color.**

B. **Adding oxalic acid ($H_2C_2O_4$) decreases the red color.**

C. **This is TRUE.**

B. **Reason (R):**

A. **Oxalic acid reacts with Fe^{3+} to form $[Fe(C_2O_4)_3]^{3-}$.**

B. **This is TRUE. $[Fe(C_2O_4)_3]^{3-}$ is a stable complex.**

49) $C_{(s)} + CO_{2(g)} \rightleftharpoons 2CO_{(g)}$

t = 0 p

t = +eq 0.5 p $2 \times 0.5p = p$

$0.5\ p + p = 12$

$p = 8$

$K_p = \frac{(8)^2}{4} = 16$ atm

50)

$A(g) + 2B(g) \rightleftharpoons 3C(g) + D(g)$

t=0 (mole) 4 7

$t=t_{eq}$ (mole) 4–x 7–2x 3x x

$\frac{[C]}{[B]} = \frac{3x}{7-2x} = 2 \Rightarrow x = 2$

$[C] = \frac{3x}{2} = \frac{3 \times 2}{2} = 3M$

51)

Based on Le Chatelier's Principle (Effect of Pressure and Temperature on Equilibrium)

52)

$K_p = \frac{P_{PQ} \times P_Q}{P_{PQ_2}}$

$K_p = \frac{200 \times 200}{400}$

$K_p = \frac{40000}{400} = 100$

$K_p = 100$
Final Answer : The value of K_p is **100**.
Hence, option (1) is correct.

53) $\underset{\text{Moles at eqm}}{} \underset{2-x}{A_2(g)} + \underset{4-x}{B_2(g)} \rightleftharpoons \underset{2x}{2AB(g)}$

$$K_c = \frac{4x^2}{(2-x)(4-x)}$$

$$\Rightarrow x = \frac{32}{24} = 1.33 \text{ mole}$$

$$[AB(g)] = \frac{2 \times 1.33}{4} = 0.66\ M$$

54) $I_2(g) \rightleftharpoons 2I(g)$
1 0
1–x 2x
at equilibrium
1–x + 2x = 1 + x
$K_C = 10^{-6}$ means concentration of reactant is greater than concentration of product.
$[I_2(g)] >>> [I(g)]$

55)

The equation provided is: log (Kp/Kc) + log RT = 0

Concept :

A. The relationship between Kp and Kc is given by: $Kp = Kc\ (RT)^{\Delta ng}$
where Δng is the change in the number of moles of gaseous species (moles of gaseous products × moles of gaseous reactants).

Calculation :
log (Kp/Kc) = – log RT log (Kp/Kc) = log $(RT)^{-1}$
Kp/Kc = $(RT)^{-1}$, Kp = Kc $(RT)^{-1}$, Kp = Kc $(RT)^{(0-1)}$ Therefore, Δng = –1

56) For conjugate acid-base at 25ºC $k_a \times k_b = 10^{-14}$, ionic product of water at 90°C is greater than 10^{-14}, for given reaction conjugate acid of NH_3 is $NH_4^{\oplus}$ & for CH_3COOH its conjugate base is $CH_3COO^{\ominus}$

57)

58) (B) $H_2PO_4^- \rightarrow HPO_4^{2-} + H^+$
acid C.B.
(C) $H_2O \rightleftharpoons H^+ + OH^-$
Temp ↑ Kw ↑

59) $Q_{sp} = [Ba^{+2}]\,[SO_4^{-2}]$
$= 10^{-4} \times [SO_4^{-2}] > 4 \times 10^{-10}$
$[SO_4^{-2}] > 4 \times 10^{-6}$ M

60) **Explain Question :** A saturated $CaCl_2$ solution gives $[Cl^-] = 2 \times 10^{-6}$ M; find solubility of $CaCl_2$ when 10^{-2} M $BaCl_2$ is present (common Cl^- ion).
Concept : This question is based on Solubility and solubility product (K_{sp}), common-ion effect
Solution : $CaCl_2 \rightleftharpoons Ca^{2+} + 2Cl^-$; if solubility = s then $[Cl^-] = 2s$ and $K_{sp} = [Ca^{2+}][Cl^-]^2$.
Common-ion (Cl^-) from $BaCl_2$ suppresses s.
From given $[Cl^-] = 2 \times 10^{-6}$ M (pure $CaCl_2$), $2s = 2 \times 10^{-6} \Rightarrow s = 1 \times 10^{-6}$ M.
$K_{sp} = [Ca^{2+}][Cl^-]^2 = s(2s)^2 = 4s^3 = 4 \times (1 \times 10^{-6})^3 = 4 \times 10^{-18}$.
In presence of 10^{-2} M $BaCl_2$, $[Cl^-] \approx 2 \times 10^{-2}$ M. Let new solubility = s'.
$K_{sp} = s'([Cl^-])^2 \approx s'(2 \times 10^{-2})^2 = s' \times 4 \times 10^{-4}$.
So $s' = K_{sp} / (4 \times 10^{-4}) = (4 \times 10^{-18})/(4 \times 10^{-4}) = 1 \times 10^{-14}$ M.
Final answer : Option (1)

61)

So acid is left at the end of reaction

$$N_{\text{final solution}} = [H^+] = \frac{N_1V_1 - N_2V_2}{V_1 + V_2}$$

$$= \frac{\frac{1}{5} \times 75 - \frac{1}{5} \times 25}{75 + 25} = \frac{1}{10} = 0.1$$

pH = –log$[H^+]$ = 1

62) $NH_4KHPO_3 \rightleftharpoons \overset{\oplus}{N}H_4 + \overset{\oplus}{K} + HPO_3^{\ominus}$
$\therefore K_{sp} = S^3$ and $S = (K_{sp})^{1/3}$

63)

Given data
Initial value of $p^H = 6$
or $[H^+] = 10^{-6}$
After 10^2 times dilution

$$H^+ = \frac{10^{-6}}{100} = 10^{-8}$$

Net $[H^+] = (10^{-8}) + (10^{-7})$ given
$= 1.1 \times 10^{-7}$ by H_2O
$p^H = 6.95$

$\Delta p^H = 6.95 - 6 = 0.95$
 In creases by 0.95 app.
Final Answer: Option_4

64)

Millimoles of $H^+ = 10^{-2} \times 200 = 2$
Millimoles of $OH^- = 10^{-2} \times 300 = 3$
$[OH^-]_{mix} = \frac{3-2}{200+300} = 2 \times 10^{-3}$
pH = 11.3

65)

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$
C – Cα Cα Cα + 0.1
≃ C Cα ≃ 0.1 M

$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$

$1.8 \times 10^{-5} = \frac{[C\alpha][0.1]}{[0.2]}$

66)

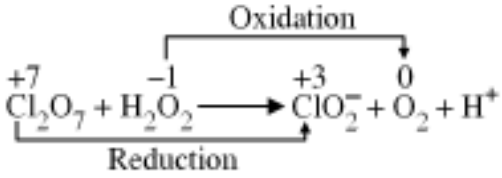
$-n_2 = -n1 + x(-1)$
 $x = n_2 - n_1$

67)

Total of oxidation number of all elements should be zero in a molecule.
ABC_2 ⇒ Total = (+1) + (+2) + 2(–1) = +1
A_2BC ⇒ Total = 2(+1) + (+2) + (–1) = +3
ABC_3 ⇒ Total = (+1) + (+2) + 3(–1) = 0
A_3BC ⇒ Total = 3(+1) + (+2) + (–1) = +4

68) $K_2 \overset{+6\times2}{Cr_2} O_7 + \overset{+n}{X^{+n}} \rightarrow \overset{+5}{X}O_3^- + 2\overset{+3}{Cr^{+3}}$
$n_f = 6$ $n_f = (5-n)$
According to law of equvalence
g equivalent of x^{+n} = g equivalent of $K_2Cr_2O_7$
$9 \times 10^{-3} \times (5 - n) = 6 \times 10^{-3} \times 6$
n = 1

69)



70)

Oxidation number of N in :-

$NH_3 \rightarrow -3$, $NO_3^- \rightarrow +5$

$NH_4^+ \rightarrow -3$, $NO_2^- \rightarrow +3$

$N_2O_5 \rightarrow +5$, $N^{-3} \rightarrow -3$

$NO \rightarrow +2$, $N_2H_4 \rightarrow -2$

$NO_2 \rightarrow +4$, $NH_2OH \rightarrow -1$

$N_2O \rightarrow +1$, $NaN_3 \rightarrow -\frac{1}{3}$

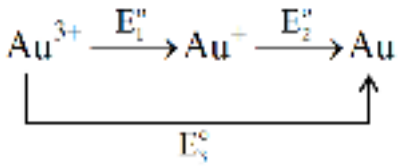
71)

Theory based

72)

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73)



$n_1E_1^o + n_2E_2^o = n_3E_3^o$

$2z + x = 3y$

74)

For 0.1 M KCl solution,

$\kappa = G \cdot G^*$

$\frac{1.29}{100} S\,cm^{-1} = \frac{1}{100\,\Omega} \times G^*$

$G^* = 1.29\ cm^{-1}$

for 0.02 M KCl solution,

$\lambda_m = G \cdot G^* \times \frac{1000}{M}$

$= \frac{1}{250} \times 1.29 \times \frac{1000}{0.02}$

$= 258\ S\ cm^2\ mol^{-1}$

$= 258 \times 10^{-4}\ S\ m^2\ mol^{-1}$

75) $\left(\frac{w}{E}\right)_{H_2} = \left(\frac{w}{E}\right)_{O_2}$

$\frac{0.5}{1} = \frac{w_{O_2}}{8}$

$w_{O_2} = 4g$

76)

Given

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell); \Delta G_f^o = -237.2 kJ/mol$

$H_2O(\ell) \rightarrow H^+(aq) + OH^-(aq); \Delta G^o = 80 kJ/mol$

At anode

$H_2(g) \rightarrow 2H^+ + 2e^-$

At cathode

$\frac{1}{2}O_2(g) + H_2O(\ell) + 2e^- \rightarrow 2OH^-$

Cell reaction

$H_2O(\ell) + H_2(g) + \frac{1}{2}O_2(g) \rightarrow 2H^+(aq) + 2OH^-(aq)$

$\Delta G° = -237.2 + (2 \times 80)$

$= -77.2$ kJ

$\Delta G^\circ = -nF E^o_{cell}$

$E^o_{cell} = -\frac{\Delta G^o}{nF} = \frac{77.2 \times 1000}{2 \times 96500} = 0.4 V$

77)

Reduction potential of copper is less than silver so tendency to gets reduced is less and hence $CuCl_2$ can be stored in silver vessel.

78)

LHS ⇒ Anode (oxidation)

$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$

RHS ⇒ Cathode (reduction)

$2Ag^+(aq) + 2e^- \rightarrow 2Ag(s)$

Net cell reaction :

$2Cl^- + 2Ag^+ \rightarrow 2Ag + Cl_2$

79)

A positive E° means that redox couple is weaker reducing agent than H^+/H_2 couple.

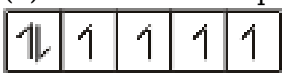
80) mol × V.F. = $\frac{Q}{F}$ $0.2 \times 1 = \frac{Q}{F}$

$Q = 0.2$ F $= 0.2 \times 96500$ C

$= 19300 = 1.93 \times 10^4$ C

81) Magnetic moment (μ) = $\sqrt{n(n+2)}$ B.M.

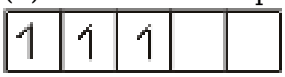
(a) $Co^{3+} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^6$



n = 4

$\mu = \sqrt{4(4+2)} = \sqrt{24}$ B. M

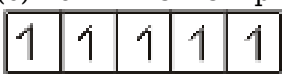
(b) $Cr^{+3} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^3$



n = 3

$\mu = \sqrt{3(3+2)} = \sqrt{15}$ B.M.

(c) $Fe^{3+} \rightarrow\ 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^5$



n = 5

$\mu = \sqrt{5(5+2)} = \sqrt{35}$ B. M.

(d) $Ni^{+2} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^8$



n = 2

$\mu = \sqrt{2(2+2)} = \sqrt{8}$ B. M.

82)

Interstitial compounds are chemically inert.

83) **Solution –** The reaction between $KMnO_4$ and oxalic acid is autocatalytic. The Mn^{2+} ions produced during the reaction act as a catalyst, increasing the reaction rate over time, leading to faster decolourisation of the purple $KMnO_4$ solution.

84) NCERT-XII, Pg. 231

85) NCERT, Pg. # 229

86)

Black phosphorus is the thermodynamically most stable allotrope of phosphorus under standard conditions. It has a layered structure, making it the least reactive and most stable form.

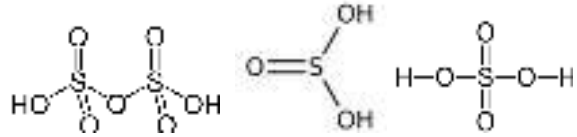
Final Answer: Option (3)

87) **Concept** - **(Enthalpy of atomization & unpaired electron)**

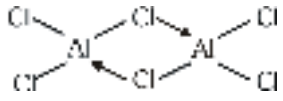
Statement-I is true; enthalpy of atomization peaks mid-series.

Statement-II is also true, as unpaired d-electrons strengthen metallic bonds. Statement-II correctly explains Statement-I's observation.

Final Answer - Option (1)

88) 

(I) (II) (III)

89)  $AlCl_3$ (dimer)

90)

NCERT Pg # 207,
ClF_3 is used for the production of UF_6 in the enrichment of U^{235}

BIOLOGY

91) NCERT-XII Pg # 69

92) NCERT XII Pg. No. 65, 66

93)

NCERT XII, page # 69

94) NCERT-XII, Pg # 54, 59

95) NCERT XII Pg. # 61

96)

Refer to module

97) Refer to module

98) NCERT-XII Pg. # 55

99) NCERT XII Pg. # 65, 66

100) Refer to module

101)

Refer to module

102)

Refer to module

103)

NCERT XII, page # 55

104)

F_1 (100%) → F_2 (50%) → F_3 (25%)

105)

NCERT XII, Pg # 64

106)

Refer to module

107)

NCERT XII, Pg. # 67

108)

Refer to module

109)

NCERT XII Pg. # 69

110)

Refer to module

111)

Refer to module

112)

Refer to module

113)

NCERT XII, Pg. # 60, 61

114)

NCERT-XI, Pg. # 249

115)

NCERT E Pg. # 318

116) B an D are correct

117)

NCERT XI, Pg. # 241, 246

118)

Allen module.

119)

NCERT XI, Pg. # 232

120)

NCERT XI, Pg. # 235, 236

121)

NCERT XI Pg.# 318

122)

NCERT XI, Pg. # 235, 236

123)

Ncert Based

124)

NCERT XI, Pg. # 232

125)

NCERT XI, Pg. # 242

126)

NCERT XI, Pg. # 245, 246

127) NCERT XI Page No. # 335

128)

NCERT XI, Pg. # 247-248

129)

The correct answer is (**3). 4**

Explanation:
'X' points to the adrenal gland (**a)**, which is also known as the suprarenal gland (**b)**. The adrenal cortex secretes mineralocorticoids (**c)**, and hyposecretion of adrenal hormones leads to Addison's disease (**d)**. Therefore, (a), (b), and (c) correctly represent 'X'.

130) NCERT Pg. # 331,333,334

131)

NCERT Pg. No 336

132)

NCERT_XI_Pg.No: 242, 243

133)

The correct answer is:
Option (1) - Hyperglycemia, Diabetes Mellitus

Explanation:

A. **Insulin** is a hormone secreted by the **pancreas** that helps lower blood sugar levels by promoting glucose uptake into cells.
B. If insulin is **malfunctioning or deficient**, glucose cannot enter the cells effectively, leading to **high blood sugar** (**hyperglycemia**).

This condition leads to **Diabetes Mellitus**, which is characterized by chronic high blood sugar.

134)

NCERT XI, Pg. # 248

135)

NCERT XI Pg. # 240

136) NCERT XIth Pg # 332, Fig. 22.2

137)

NCERT Page # 42-43

138)

NCERT XI Pg. # 28

139)

NCERT XI Pg. # 29

140) NCERT XI, Pg. # 38

141)

NCERT XI Pg. # 34

142)

NCERT XI Pg. # 30

143)

NCERT XI Pg. # 30

144)

NCERT XI Pg. # 33

145) **Explain Question :** The question asks you to identify the four labeled parts A, B,C and D in the provided diagram of a moss plant
Concept: This question is based on Mosses
Explanation : A : capsule B: seta C : leaves D: rhizoids

Final answer : 4

146) NCERT-XI Pg. # 39, 31, 37, 34

147)

NCERT XI Pg. # 33

148)

NCERT XI Pg. # 25 Fig. 3.1

149)

NCERT XI Pg. # 28 Fig. 3.2
Explain Question : The question has provided a diagram of a moss plant and its life cycle stages, labeled A, B, C, and D. The task is to correctly identify each labeled part and match it to one of the four given options.
Concept : This question is based on life cycle of Bryophytes - moss.
Explanation :
A: Capsule- This is the spore-producing part of the sporophyte, located at the apex.
B: Seta- This is the stalk that supports the capsule, connecting it to the gametophyte.
C: Sporophyte- This refers to the entire diploid structure, which includes the foot (embedded in the gametophyte), the seta, and the capsule.
D: Gametophyte- This represents the entire haploid plant body, which is the dominant stage in the moss life cycle and includes the leaves, main axis, and rhizoids.

Final answer : option (2)

150) **Explain Question:** Question asking about the identify given diagram and select correct statement according to given diagram.
Concept: This question is based on Pteridophyta
Solution: The given diagram is of *Salvinia.*
(C) It produce two different type of spore. **Correct because *Salvinia* is a heterosporous pteridophyte**
(D) The development of the zygote into young embryo takes place within the female gametophyte.
Correct for heterosporous pteridophyte Salvinia also heterosporous
Final answer: option (3)